AI Study Mentor

Unit 22: Application Development — Assignment Part 2

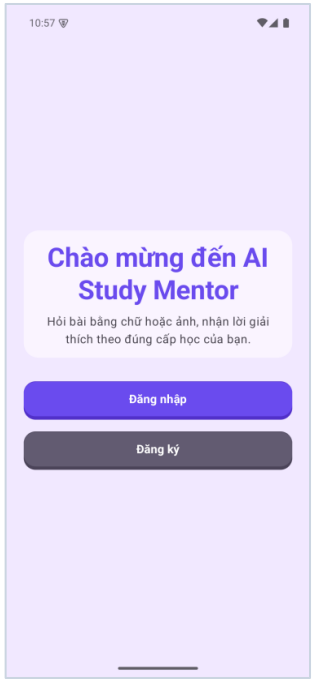
Nguyen Minh Khang (BC00720) · SE08201
Assessor: Mr. Le Nhat Quang
Group/member fields: N/A — Individual submission
Presentation date: [PENDING]

DRAFT — NOT FOR FINAL SUBMISSION

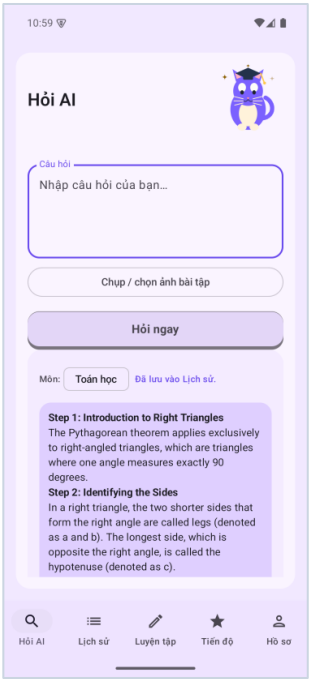
CONTEXT

The problem: help that is generic, unverified and lost

- Students working alone receive explanations that are hard to check and not tuned to their level.
- Answers disappear after the session, so progress cannot be seen.
- Target users: middle school, high school and university students; BrightPath owns the product.
- Constraints: junior team, 14-week MVP, limited budget and an API cost ceiling.
  - Design response: local-first storage, a cache in front of the provider, structured answers and a testable domain tier.



real-final/01-welcome.png



real-final/06-ask-gemini-answer.png

FINAL evidence — real Gemini, fixed APK

Requirements and priorities

Band	Requirements	Why
MVP	FR1 auth, FR2 onboarding, FR3 text/image ask, FR4 structured answer, FR6 history, FR8 progress	Core study loop
Post-MVP	FR5 improvement blocks, FR7 practice, FR9 gamification, FR10 notifications	Scheduled after the loop
NFR	NFR1 responsiveness, NFR2 three actions, NFR3 API 33+, NFR4 encryption, NFR5 offline reading, NFR6 auth hardening	Quality constraints

Deferred items are scheduled work, not failures.

Proposed solution and the frozen Part 1 design

- Kotlin + Jetpack Compose, single activity, one module.
- Room for local-first storage; manual dependency graph instead of a framework.
- Gemini called through HttpURLConnection and org.json.
- Ask path: normalise → exact hash → similarity policy → provider → persist → history.
- Cache before AI is the cost-control mechanism the brief requires.

4

layers: ui, domain, data, platform

98

Kotlin files, 12,103 lines

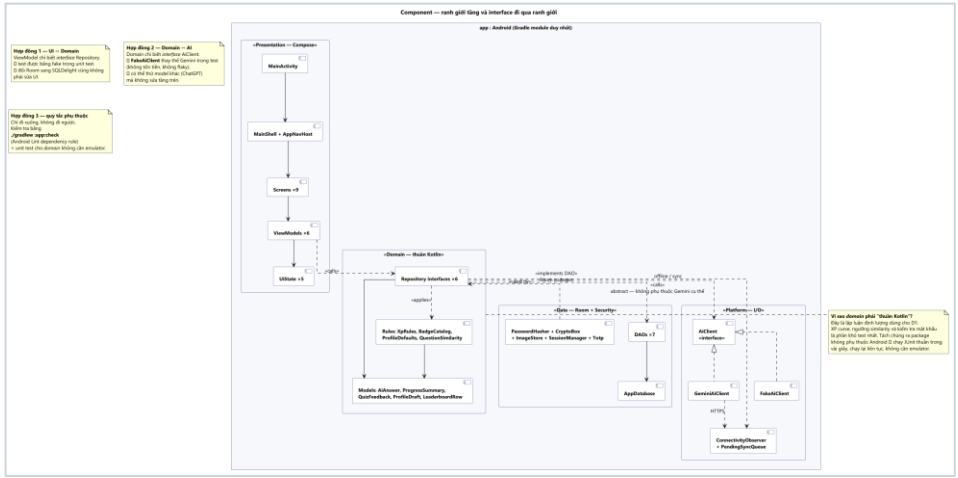
Design decisions D-1…D-12 are the M4 baseline.

Development method and toolchain

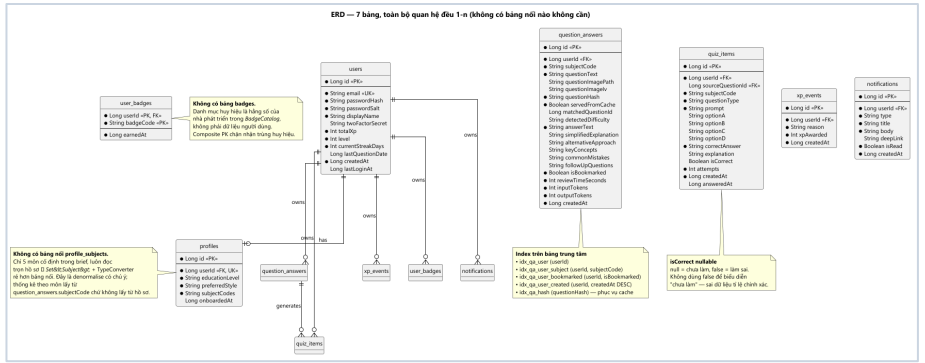
- One-week increments; Definition of Done per phase.
- Rule: every logic change ships with a test; tests never call the real provider.
- AGP 9.3.3 · Kotlin 2.2.10 · Compose BOM 2026.02.01 · Room 2.7.2 · KSP 2.3.12.
- CI runs ./gradlew test with no API key, so it builds against the offline fake client.
- Thirteen PlantUML sources are kept as text and reviewed as diffs.
  - Honest record: 10 of 14 sprint entries are empty; 13 commits on the audited branch, single author.

As-built architecture and data flow

- Four layers: ui, domain, data, platform; one module.
- Room schema v1, seven tables: users, profiles, question_answers, quiz_items, xp_events, user_badges, notifications.
- Ask flow: normalise → hash → similarity → provider → persist → history.
- A cache hit costs no API call; a number or language difference forces a fresh call.

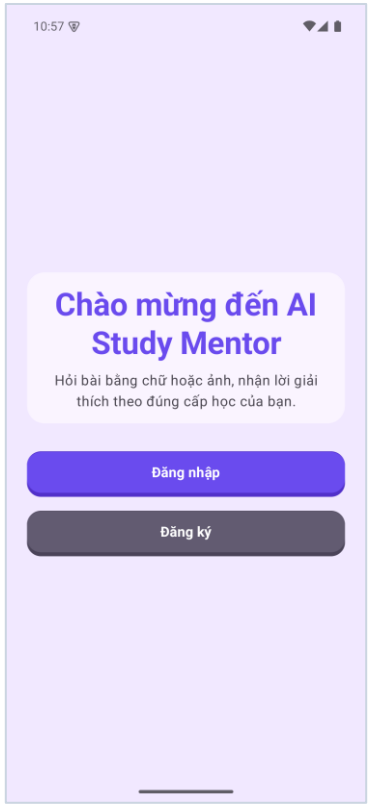


Component view — docs/uml/png/06_component.png

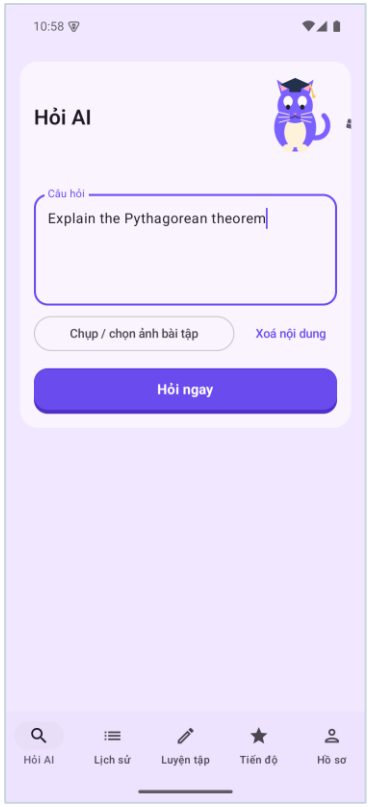


ERD, seven tables — docs/uml/png/09_erd.png

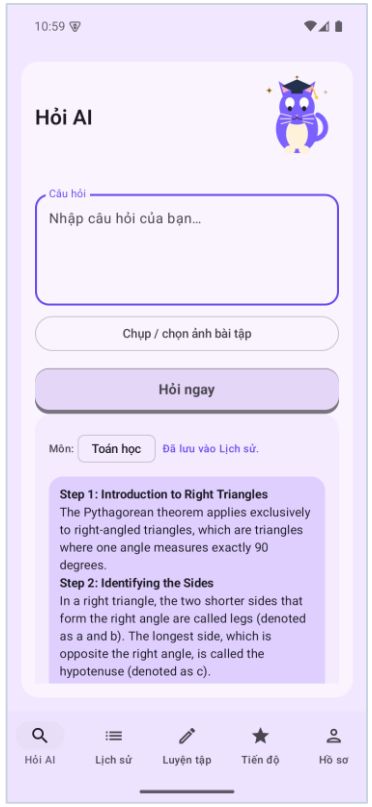
Demo 1: a real question, a real Gemini answer



01 · welcome



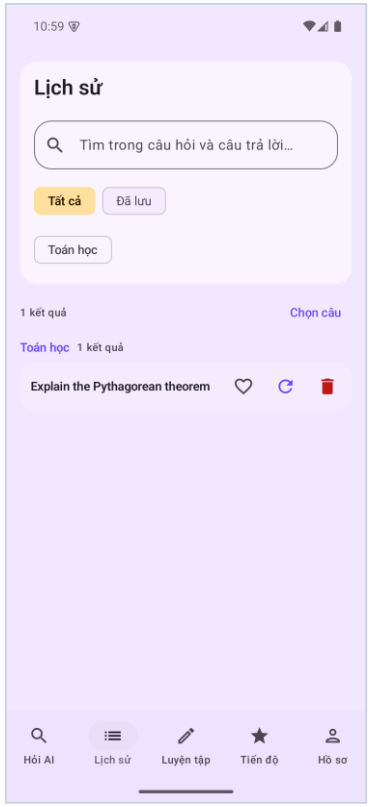
05 · question



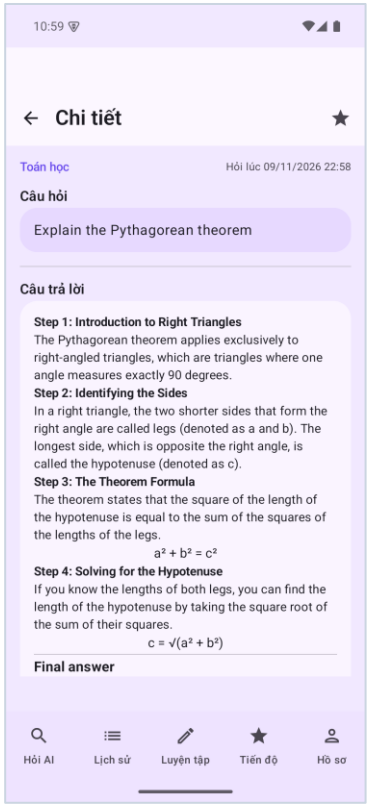
06 · real answer

- POST …:generateContent → HTTP 200, modelVersion gemini-3.5-flash-lite.
- Answer shows steps and is saved to history.
- One real response evidences the path; it does not establish answer quality.

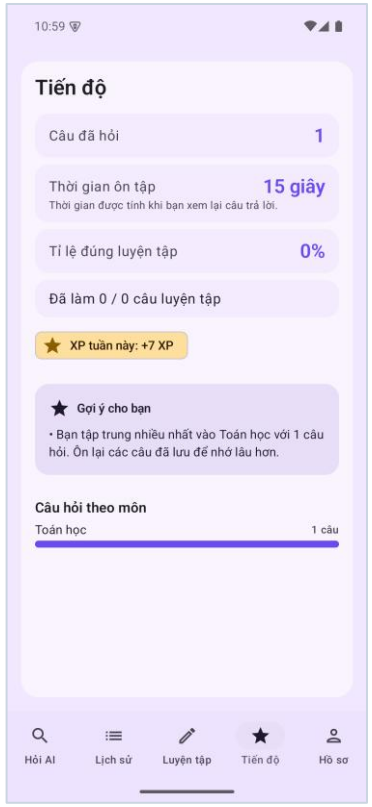
Demo 2: history, detail and progress



07 · history



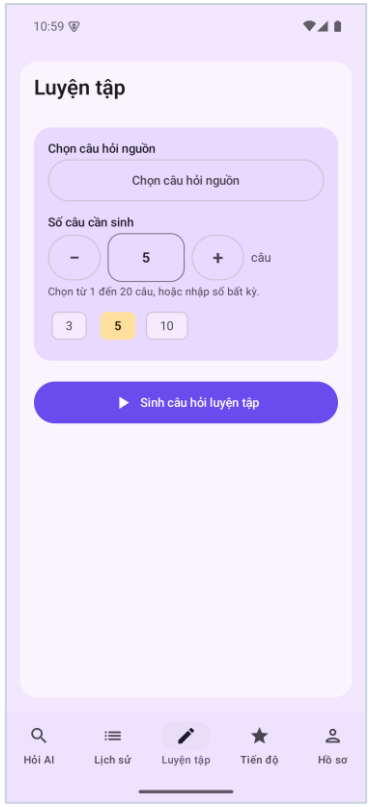
08 · detail



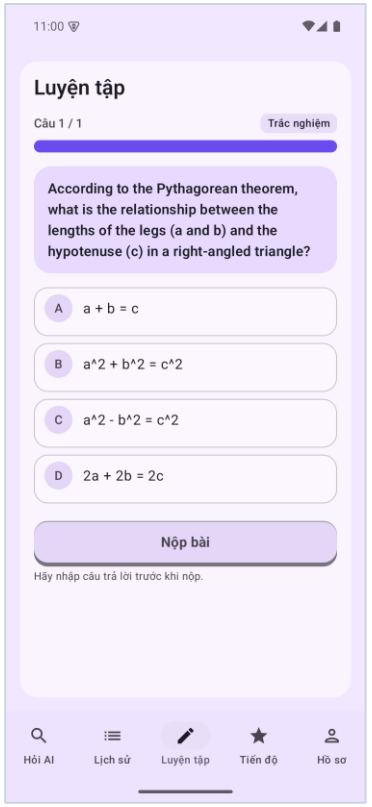
09 · progress

- Progress observed: 1 question, 15 s, 0%, +7 XP.
- Search, bookmark and subject filters were not exercised.

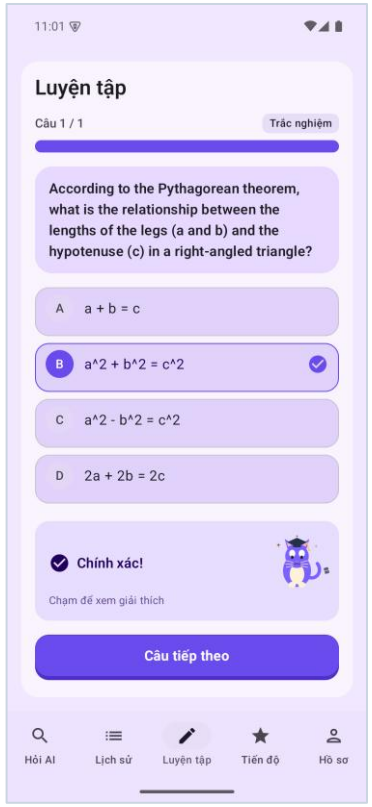
Demo 3: practice and corrected feedback



10 · setup



11 · quiz



12 · corrected

- Letter/number keys now resolve to option text before comparison.
- Seven PracticeAnswerTest regressions; screenshot 12 shows “Chính xác!”.
- The screenshot used an option-text key; the letter-key path is unit-tested.

Evidence and testing

159
JVM tests, 13 classes, 0 failures

39/39
instrumented tests after the fix

16
real-final screenshots

- NFR1 cold start: 3041 / 3108 / 3284 / 3310 / 5139 ms — median 3284, n=5.
- Gemini Ask round trip: 861 / 913 / 2207 ms — median 913, 3/3 HTTP 200.
- Observations only: no project SLO is defined.
- Tests prove logic and data behaviour, not AI accuracy.

Evaluation against requirements

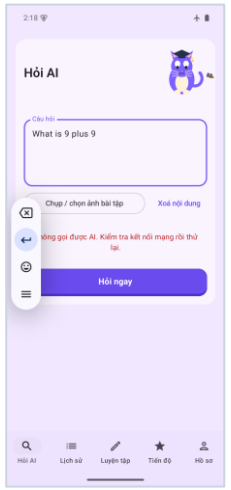
Status	Requirements	Basis
Met	FR1, FR4 (one response), FR6 list/detail, FR7 generation/feedback, FR8	Real-final runtime
Partial	FR2 subjects, FR3 image, FR9 leaderboard, FR10 reminders	Observed gaps
Unverified	FR5 improvement blocks	Not exercised
NFR	NFR1 measured observation; NFR3 met; NFR4–NFR6 partial	NFR2 and history timing pending

No row is rounded up; partial stays partial.

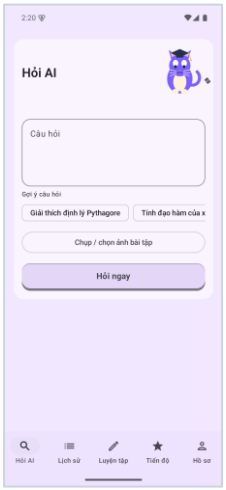
Critical review, feedback and next steps

- Strengths: testable domain tier, disclosed trade-offs, measured cost control.
- Weaknesses: onboarding subjects, image lifecycle, inactive abuse filter, subject mapping.
- Risks: hallucinated content, cost ceiling, privacy scope, sprint-record gaps.
- Peer feedback and decisions: [PENDING: survey responses].
- Backlog: 2FA, image lifecycle, offline queue, abuse filter, server-backed leaderboard.

Supplementary evidence — not part of the 16-image final gate



offline error



login after logout

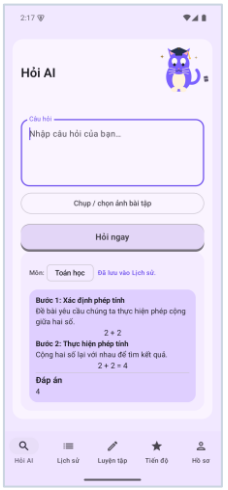


image answer